SRS · REQUIREMENTS SPECIFICATION

Software Requirements Specification

Smart Inventory and Procurement System

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A simple web system that manages inventory, procurement, sales, and finance (invoices and payments) for a hypothetical multi-branch company, plus three smart features powered by Gemini that warn management before problems occur.

1. Project Overview and Scope

The system tracks a product through the company's internal supply chain in four stages:

1. **Supplier:** the product is purchased from the supplier through a purchase order.
2. **Warehouse:** the product is received and its stock balance is kept here.
3. **Branch:** each branch has a warehouse, and products are transferred to it when needed.
4. **Sale:** the sale is recorded at the branch and deducted from its warehouse stock.

Alongside the product flow, the system tracks the money that follows it: the supplier invoice and the payment to the supplier on the purchasing side, and the sales invoice and the payment received on the sales side.

System goals

- An accurate, up-to-date stock balance for every product in every warehouse.
- A clear purchasing cycle from purchase order to receipt.
- Automatic stock deduction on every sale.
- Smart recommendations that come before shortages or dead stock occur.
- A sales invoice and a recorded payment for every sale.
- Supplier invoices tracked from receipt until fully paid, with overdue invoices flagged.
- Simple financial reports: sales, gross profit, supplier payables, and stock value.

> **Out of scope:** a full general ledger (double-entry bookkeeping), payroll, general expenses, tax filing, and bank or online payment gateway integration; POS and barcode devices; customer management (CRM) and customer credit accounts; shipment tracking; manufacturing; a supplier portal; and a mobile app.

2. Assumptions

- The system serves one company with one currency and about 3 branches.
- Each branch has one warehouse. If a product runs out in one warehouse, it is transferred from another branch's warehouse.
- New branches and warehouses can be added from inside the system without changing code.
- The product code (SKU) is entered inside the system, with no barcode device integration.
- Sales are paid in full at the time of sale (cash, card, or bank transfer); there are no customer credit accounts.
- Purchases from suppliers are on credit: the supplier sends an invoice with a due date, and the company pays it in full or in parts.
- Payments are recorded manually in the system; there is no connection to banks or online payment gateways.
- A single tax rate (which can be set to 0%) is configured in the system and applied to sales invoices.
- The accounting is simple: gross profit is sales revenue minus the purchase cost of the goods sold.
- Historical sales data is sample data, generated for demonstration only.
- The smart features use a low-cost Gemini model with a System Prompt, while all numbers are calculated inside the system itself.

3. Actors and Permissions

The system has six actors, and each actor sees only their own screens.

Actor	Responsibility	Available screens
Admin	Manages users, branches, and warehouses	All screens
Storekeeper	Stock movements, transfers, stocktaking, and goods receipt	Inventory, Transfers, Stocktaking, Receiving
Purchasing Officer	Manages suppliers and purchase orders	Suppliers, Purchase Orders, Reorder Recommendations
Accountant	Issues and reviews invoices, records supplier payments, and prepares financial reports	Sales Invoices, Supplier Invoices, Payments, Financial Reports
Branch Manager	Monitors branch stock and sales	Reports, Forecasts, Dead Stock
Sales Clerk	Records sales and the payments received	Sales screen (with invoice printing)

4. Functional Requirements

Each ID is FR (functional requirement), a module abbreviation, and a number. It is used to trace every requirement to the use case that covers it.

ID	Module	Requirement
FR-USR-01	Users	Log in, and see only the screens allowed for the user's role.
FR-USR-02	Users	The Admin creates users and assigns each one a role.
FR-BR-01	Branches and Warehouses	Add a branch and a warehouse linked to it, with one warehouse per branch.
FR-BR-02	Branches and Warehouses	View branches and their warehouses.
FR-PRD-01	Products	Add a product with name, SKU, category, purchase price, selling price, minimum stock level, and safety stock.
FR-PRD-02	Products	Edit and view products, and prevent duplicate SKUs.
FR-INV-01	Inventory	View the stock balance of each product in each warehouse.
FR-INV-02	Inventory	Record a stock movement (in, out, adjustment), update the balance, and keep a movement history.
FR-INV-03	Inventory	Transfer a quantity from one warehouse to another; the transfer is rejected if it exceeds the available balance.
FR-INV-04	Inventory	Perform a stocktake: compare actual quantities with recorded ones and settle the difference with an adjustment movement.
FR-SUP-01	Suppliers	Add a supplier with the products they supply, their prices, and the usual lead time.
FR-PO-01	Purchase Orders	Create a purchase order with the supplier, products, quantities, and receiving warehouse.
FR-PO-02	Purchase Orders	Move the order through the stages: requested, sent, partially received, fully received.
FR-PO-03	Purchase Orders	Record receipt with the actual date and the received quantity against the ordered quantity, and create a stock-in movement automatically.
FR-SAL-01	Sales	Record a sale with branch, product, quantity, and date.
FR-SAL-02	Sales	Deduct the quantity automatically from the branch warehouse, and reject the sale if it exceeds the balance.

ID	Module	Requirement
FR-SIN-01	Sales Invoices	Generate a sales invoice automatically for every sale, with a unique sequential number, branch, date, items, quantities, unit prices, discount, tax, and total.
FR-SIN-02	Sales Invoices	View and search sales invoices by number, date, and branch, and print an invoice as a PDF.
FR-SIN-03	Sales Invoices	Cancel a sales invoice with a reason; the quantity returns to the branch warehouse, and the invoice is kept with a cancelled status (invoices are never deleted).
FR-PIN-01	Supplier Invoices	Record a supplier invoice linked to a purchase order, with invoice number, invoice date, due date, and amount; prevent a duplicate invoice number for the same supplier.
FR-PIN-02	Supplier Invoices	Compare the invoice amount with the value of the goods actually received (received quantity × agreed price) and flag any difference.
FR-PIN-03	Supplier Invoices	Track the status of each supplier invoice: unpaid, partially paid, paid, or overdue (past its due date and not fully paid).
FR-PAY-01	Payments	Record the payment for each sale with its method (cash, card, or bank transfer); the amount equals the invoice total and the invoice is marked paid.
FR-PAY-02	Payments	Record a payment to a supplier against a supplier invoice with method, amount, and date; allow partial payments and reject a payment larger than the remaining balance.
FR-PAY-03	Payments	Show the payment history and the remaining balance of each supplier invoice.
FR-RPT-01	Reports	Current stock report.
FR-RPT-02	Reports	Shortage report: products below the minimum stock level.
FR-RPT-03	Reports	Best-selling and slowest-selling products report.
FR-FIN-01	Financial Reports	Sales report: total sales and tax by branch, product, and period.
FR-FIN-02	Financial Reports	Gross profit report: sales revenue minus the purchase cost of the goods sold, by branch, product, and period.
FR-FIN-03	Financial Reports	Supplier payables report: outstanding supplier invoices by supplier and due date, with overdue invoices highlighted.
FR-FIN-04	Financial Reports	Daily collections report: payments received per branch, split by payment method.

ID	Module	Requirement
FR-FIN-05	Financial Reports	Stock valuation report: stock balance multiplied by purchase price, per warehouse and in total.
FR-FIN-06	Financial Reports	Export any financial report to CSV.

5. Smart Features

The split of work is clear: the system calculates the numbers and prepares a data summary, then sends it to Gemini with a fixed System Prompt that defines its role. Gemini returns a short, structured recommendation, which the system shows to the user.

A. Smart Reorder

The reorder point of each product is calculated with this formula:

REORDER POINT FORMULA

Reorder Point = (Avg Daily Sales × Lead Time) + Safety Stock

Avg Daily Sales × Lead Time: Expected sales during the supplier's lead time (daily average × days)

Safety Stock: Buffer kept for unexpected demand

When the stock balance falls below it, the system sends the product data (balance, average sales, lead time) to Gemini, which suggests an order quantity with a short reason. This appears as an alert for the Purchasing Officer.

B. Demand Forecasting

The system aggregates monthly sales per product and calculates a moving average as a baseline. It sends both to Gemini, which forecasts next month's sales with a brief explanation of the trend.

C. Dead Stock Detection

The system finds products that have not sold for X days (a configurable value) and sends them to Gemini, which suggests for each one either a discount or a transfer to another branch.

ID	Requirement
FR-AI-01	Calculate the reorder point for each product and show an alert when the balance falls below it.
FR-AI-02	Get a suggested order quantity and its reason from Gemini, and show them with the alert.
FR-AI-03	Forecast each product's sales for next month and show the forecast with its explanation on screen.

ID	Requirement
FR-AI-04	Identify products with no sales in X days, and suggest a discount or a transfer for each.
FR-AI-05	Use a fixed System Prompt that defines Gemini's role and requires a structured response; Gemini is not used for numeric calculations.

6. Non-Functional Requirements

Section 4 and Section 5 describe **what the system does** (functional requirements). This section describes **how well the system does it**: speed, security, reliability, ease of use, and similar qualities. A functional requirement is a feature the user can see and use (for example, recording a sale). A non-functional requirement is a quality or constraint on that feature (for example, recording a sale must take less than 2 seconds). Each ID is NFR (non-functional requirement), a category abbreviation, and a number.

	Functional (FR)	Non-Functional (NFR)
Answers	What does the system do?	How well does it do it?
Example	The system deducts stock on every sale.	The deduction completes in under 2 seconds.
Testing	Works or does not work.	Measured against a target value.

6.1 Performance

ID	Category	Requirement
NFR-PER-01	Response time	Ordinary screens (lists, forms, stock balance) load in 3 seconds or less under normal load.
NFR-PER-02	Response time	Recording a sale (with its invoice and payment) or a stock movement completes, with the balance updated, in 2 seconds or less.
NFR-PER-03	Reports	Stock, shortage, sales, and financial reports are generated in 5 seconds or less for the sample data volume.
NFR-PER-04	Capacity	The system supports at least 20 concurrent users across all branches without noticeable slowdown.

6.2 Security

ID	Category	Requirement
NFR-SEC-01	Authentication	Every screen except login requires an authenticated session; the session expires after 30 minutes of inactivity.
NFR-SEC-02	Passwords	Passwords are stored only as salted hashes (never as plain text) and have a minimum length of 8 characters.
NFR-SEC-03	Authorization	Role permissions (Section 3) are enforced on the server for every request, not only by hiding screens.
NFR-SEC-04	Transport	All traffic between the browser and the server uses HTTPS.
NFR-SEC-05	Audit	Stock movements, sales, invoices, payments, and purchase order changes record the user and the time of the action.
NFR-SEC-06	Secrets	The Gemini API key is kept on the server in configuration, never in code or in the browser.
NFR-SEC-07	Financial records	Issued invoices and recorded payments cannot be edited or deleted; a mistake is corrected by cancelling the invoice, and the cancellation is recorded with its reason and user.
NFR-SEC-08	Financial access	Financial screens and reports are limited to the Accountant and the Admin (Section 3).

6.3 Reliability and Data Integrity

ID	Category	Requirement
NFR-REL-01	Atomic operations	A sale (with its invoice and payment), transfer, or goods receipt either completes fully or not at all; the stock balance is never left half-updated.
NFR-REL-02	Concurrency	If two users sell the last units at the same time, only one succeeds and the stock balance never becomes negative.
NFR-REL-03	Traceability	The stock balance always equals the sum of its movement history, so any figure can be explained.
NFR-REL-04	Backup	The database is backed up daily, and the backup can be restored.
NFR-REL-05	Availability	The system is available at least 99% of the time during working hours.
NFR-REL-06	Invoice numbering	Invoice numbers are unique and sequential with no duplicates, even when several users issue invoices at the same time.

ID	Category	Requirement
NFR-REL-07	Money accuracy	Money amounts are stored as exact decimal values (never floating point) and rounded to 2 decimals in one consistent way.
NFR-REL-08	Balance consistency	An invoice total always equals the sum of its lines plus tax, and a supplier invoice balance always equals its amount minus its payments.

6.4 Usability

ID	Category	Requirement
NFR-USA-01	Ease of learning	A new user can complete the main task of their role (for example, a Sales Clerk recording a sale) after 15 minutes of training.
NFR-USA-02	Feedback	Every action shows a clear success or error message; a rejected operation (such as selling more than the balance) states the reason.
NFR-USA-03	Input checking	Forms validate input (required fields, positive quantities, valid dates) before saving.
NFR-USA-04	Consistency	All screens share one layout and one terminology for products, warehouses, and orders.
NFR-USA-05	Invoice printing	An invoice prints on A4 and shows the company, branch, invoice number, date, items, and totals.

6.5 Compatibility and Maintainability

ID	Category	Requirement
NFR-CMP-01	Browsers	The system works on the latest versions of Chrome, Edge, and Firefox, with no installation on the user's device.
NFR-CMP-02	Screen size	Screens are usable on desktop and tablet screen sizes.
NFR-MNT-01	Configurability	Branches, warehouses, the dead stock threshold (X days), the tax rate, and the payment methods are changed from inside the system without changing code.
NFR-MNT-02	Scalability	The design allows adding branches and products without redesigning the database.
NFR-MNT-03	Code quality	The code is organized by module (Section 4) and documented well enough for a new developer to extend it.

6.6 Smart Features (Gemini)

ID	Category	Requirement
NFR-AI-01	Response time	A Gemini recommendation is shown within 10 seconds; after that the request times out.
NFR-AI-02	Fallback	If Gemini is unavailable or times out, the system still shows the calculated numbers (reorder point, moving average, idle days) with a message that no recommendation is available. Core inventory, purchasing, sales, and finance functions are not affected.
NFR-AI-03	Output format	The response follows the structure required by the System Prompt; an invalid response is rejected and handled as in NFR-AI-02.
NFR-AI-04	Data privacy	Only product and sales summaries are sent to Gemini; no user names, passwords, invoice or payment details, or other personal data.
NFR-AI-05	Cost control	Requests are sent only when an alert or report is opened or refreshed, and results are cached to limit the number of calls to the low-cost model.

Note: the target values above are reasonable starting points for a demonstration system of about 3 branches; they can be adjusted with the supervisor or client.